



Estimating the Relationship Between Drug Use and Major Depressive Episodes Among Youth

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Introduction

This study estimates the relationship between drug use history and lifetime prevalence of a Major Depressive Episode (MDE) among US adolescents aged 12-17, utilizing data from the National Survey on Drug Use and Health (NSDUH). Adolescence is a critical period characterized by increased risk for substance use initiation and associated adverse outcomes.¹ I aim to disaggregate drug use into cultural categories—“soft” versus “hard” drugs—to determine if the type of substance use differentially predicts MDE status in adolescents.

Methods

This study employs a cross-sectional, secondary data analysis of the NSDUH public-use data file. The sample is restricted to youth only, defined as respondents aged 12-17 who completed the survey’s adolescent section. The primary outcome variable is the self-reported, lifetime MDE, as defined by the NSDUH based on DSM-5 criteria. The primary predictor variables are constructed from lifetime “ever tried” responses for specific substances, categorized as:

- **“Soft” Drugs:** Alcohol, Nicotine (cigarettes, cigars, vaping, smokeless tobacco, etc.), Marijuana and CBD/Hemp.
- **“Hard” Drugs:** Cocaine, Crack, Methamphetamine, Heroin, Hallucinogenics, Inhalants, and non-medical use of prescription drugs (e.g. pain relievers, tranquilizers).
- **Any Drug Use:** A binary variable indicating the use of any substance in either category.

Logistic regression models were used to estimate the odds ratios (OR) for the association between each drug use category (any, soft, and hard) and the likelihood of reporting a lifetime MDE, controlling for relevant confounding variables.

References

1. Gray KM, Squeglia LM. Research Review: What have we learned about adolescent substance use? J Child Psychol Psychiatry. 2018 Jun;59(6):618-627. doi: 10.1111/jcpp.12783. Epub 2017 Jul 17. PMID: 28714184; PMCID: PMC5771977.

Results

Table 1: Characteristics of sample population.

Characteristics	Unweighted (N = 8,427)		Weighted (N = 19,200,774)		95% Confidence Interval
	n	%	n	%	
Sex	Male	4,250	50.4%	9,655,581	50.3%
	Female	4,177	49.6%	9,545,193	49.7%
Age Group (years)	12-13	2,468	29.3%	5,456,387	28.4%
	14-15	3,054	36.2%	6,860,224	35.7%
	16-17	2,905	34.5%	6,884,163	35.9%
Race/Ethnicity	NonHispanic White	3,765	44.7%	9,609,558	50.0%
	NonHispanic Black/African American	1,060	12.6%	2,559,267	13.3%
	NonHispanic Native American/Alaska Native	263	3.1%	140,696	0.7%
	NonHispanic Native Hawaiian/Other Pacific Islander	36	0.4%	53,997	0.3%
	NonHispanic Asian	421	5.0%	1,134,541	5.9%
	NonHispanic more than one race	641	7.6%	686,367	3.6%
	Hispanic	2,241	26.6%	5,016,348	26.1%
Education	5th grade or lower	17	0.2%	23,085	0.1%
	6th grade	367	4.4%	695,183	3.6%
	7th grade	1,158	13.7%	2,574,345	13.4%
	8th grade	1,375	16.3%	3,121,298	16.3%
	9th grade	1,538	18.3%	3,558,964	18.5%
	10th grade	1,552	18.4%	3,450,451	18.0%
	11th grade	1,500	17.8%	3,548,809	18.5%
	12th grade	856	10.2%	2,091,764	10.9%
	College or university/1st year	61	0.7%	119,484	0.6%
	College or university/2nd year, 3rd year	3	0.0%	17,390	0.1%
Poverty Level	Living in Poverty	1,933	22.9%	3,960,665	20.6%
	Income Up to 2X Fed Pov Thresh	1,782	21.1%	3,847,436	20.0%
	Income More Than 2X Fed Pov Thresh	4,712	55.9%	11,392,672	59.3%
Average Grades	D or Lower	475	5.6%	839,876	4.4%
	A, B, or C	7,952	94.4%	18,360,897	95.6%
Parental Monitoring on School Nights	Always/Sometimes	4,874	57.8%	11,128,954	58.0%
	Seldom/Never	3,553	42.2%	8,071,819	42.0%

Table 1. Unweighted and weighted demographics and characteristics of sample population.

Table 3: Logistic Regression Models for Any Drugs Ever Used.

	OR	95% CI
Crude Model		
Ever Used	2.86	(2.48, 3.31)
Model 1 - Biological Epoch		
Ever Used	2.64	(2.26, 3.08)
Model 2 - Social Epoch		
Ever Used	2.58	(2.20, 3.02)
Model 3 - Behavioral Epoch		
Ever Used	2.57	(2.19, 3.01)

Table 3. Logistic regression models for any drugs ever used. Model 1 - biological epoch adjusts for age group and sex. Model 2 - social epoch adjusts for race and current grade level. Model 3 - behavioral epoch adjusts for arguments with parents.

Results Continued...

Table 4: Logistic Regression Models for “Soft” Drugs Ever Used.

	OR	95% CI
Crude Model		
Ever Used	2.93	(2.48, 3.46)
Model 1 - Biological Epoch		
Ever Used	2.57	(2.13, 3.08)
Model 2 - Social Epoch		
Ever Used	2.62	(2.14, 3.22)
Model 3 - Behavioral Epoch		
Ever Used	2.61	(2.13, 3.20)

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Table 4. Logistic regression models for “soft” drugs ever used. Model 1 - biological epoch adjusts for age group and sex. Model 2 - social epoch adjusts for race and current grade level. Model 3 - behavioral epoch adjusts for arguments with parents.

Table 5: Logistic Regression Models for “Hard” Drugs Ever Used.

	OR	95% CI
Crude Model		
Ever Used	2.21	(1.91, 2.57)
Model 1 - Biological Epoch		
Ever Used	2.09	(1.80, 2.42)
Model 2 - Social Epoch		
Ever Used	2.03	(1.73, 2.39)
Model 3 - Behavioral Epoch		
Ever Used	2.04	(1.73, 2.40)

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Table 5. Logistic regression models for “hard” drugs ever used. Model 1 - biological epoch adjusts for age group. Model 2 - social epoch adjusts for race and current grade level. Model 3 - behavioral epoch adjusts for arguments with parents.

Conclusion

Results were consistent with existing literature: drug use is significantly associated with increased odds of having a MDE, but odds were not different between “soft” and “hard” drug use. The adjusted odds ratios for any, “soft”, and “hard” drug use were 2.57, 2.61, and 2.04, respectively.

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